

Lecture Notes: Graph Theory, Kruskal's Algorithm

Math F113X Math and Society

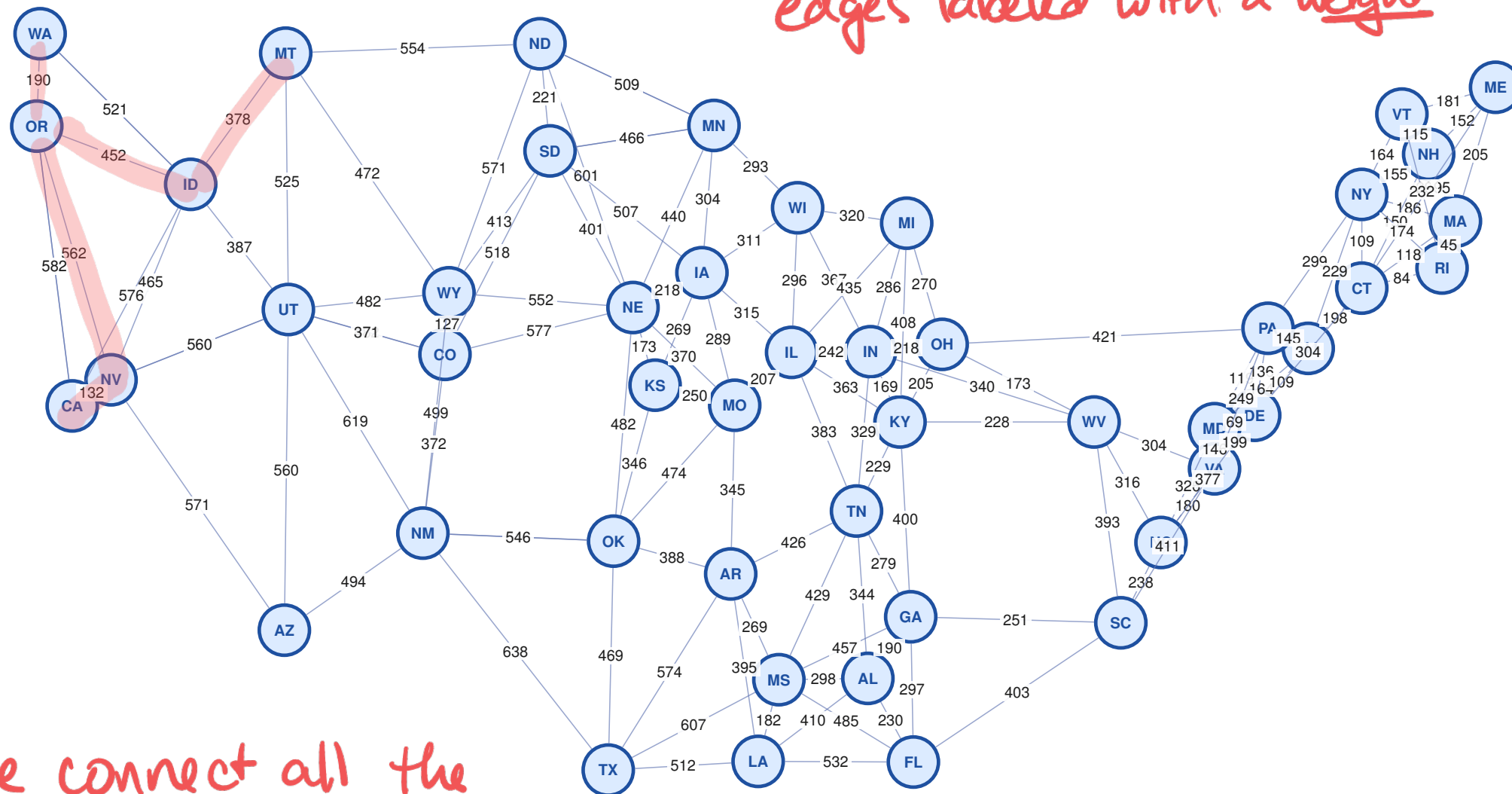
Goals:

- Understand the terms: tree, spanning tree, minimum cost spanning tree
- Understand how to use Kruskal's Algorithm to find a minimum cost spanning tree
- Know of applications of minimum cost spanning trees

48 contiguous US state capitals

Edge weights = approximate road distance in miles. Each capital connected to its 4–5 nearest neighbors.

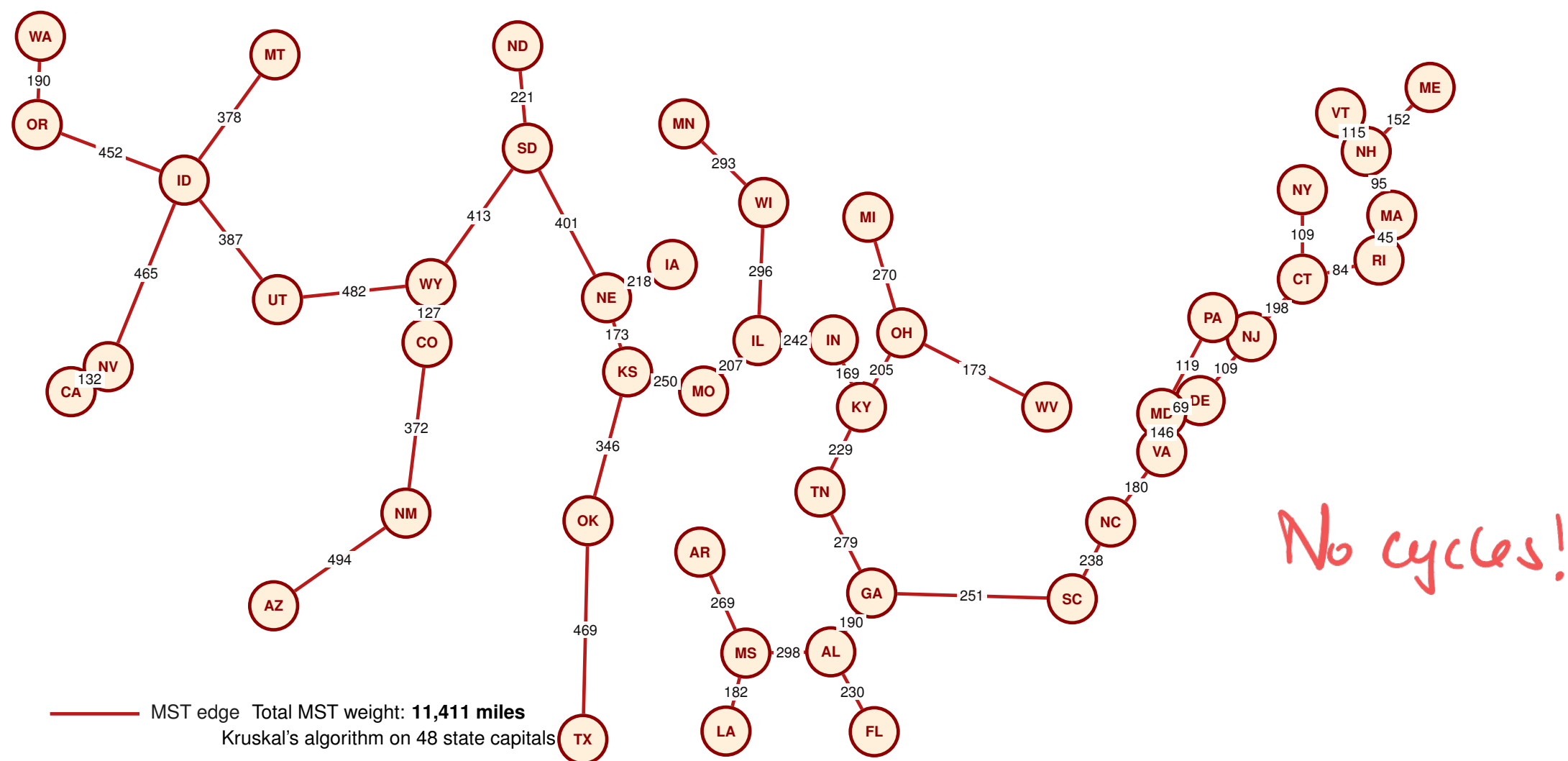
*vertices = capitals/states, edges = nearest neighbors
edges labeled with a weight*



Can we connect all the capitals & minimize the total road distance?

Minimum Weight Spanning Tree

Total MST weight $\approx 11,411$ miles. Red edges form the unique spanning tree of minimum total road distance.



Definitions

(a) **weighted graph** A graph with weights on the edges

ex: cost, distance, capacity,
time...

(b) **tree**

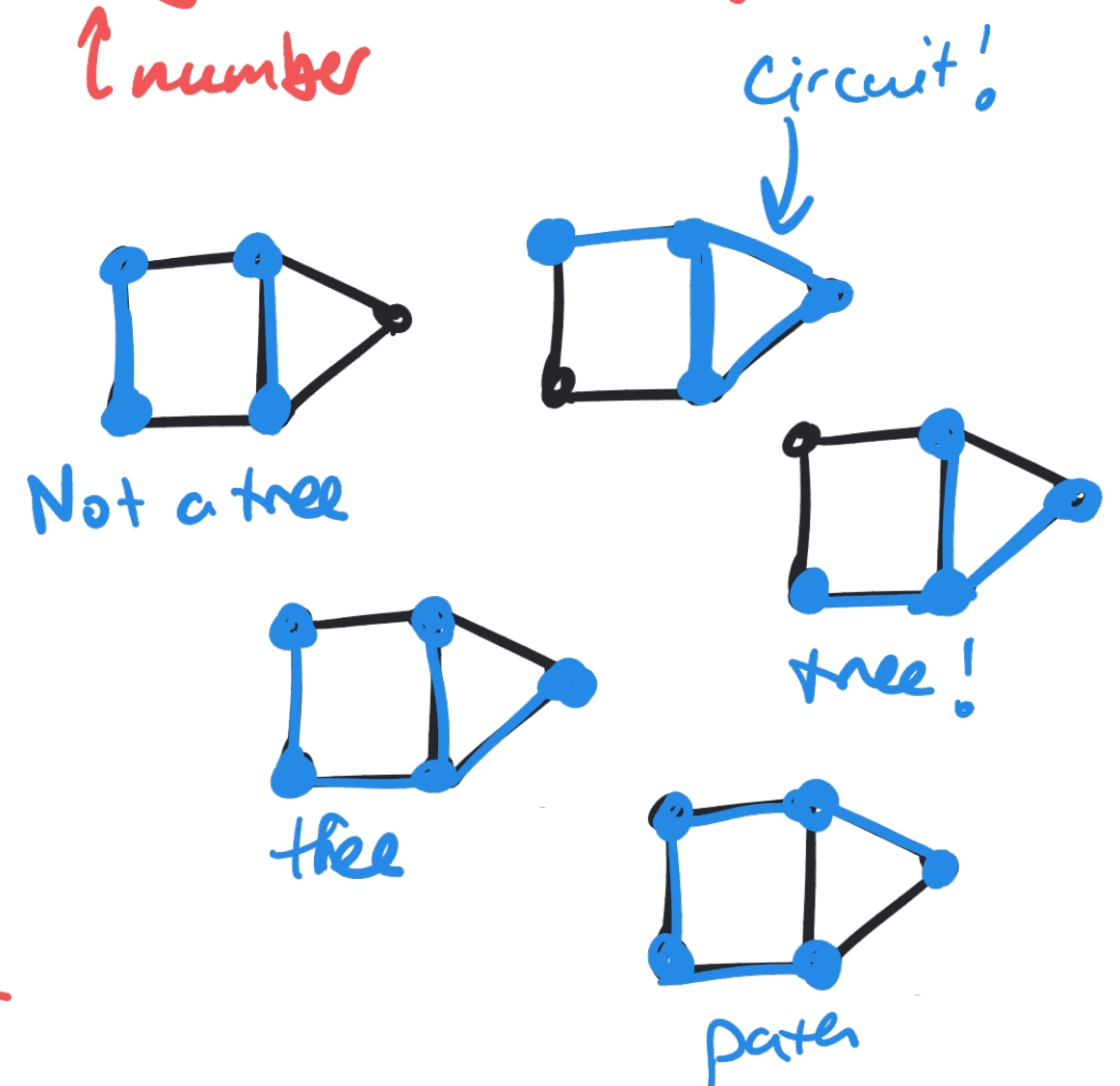
Connected graph with no circuits

(c) **spanning tree**

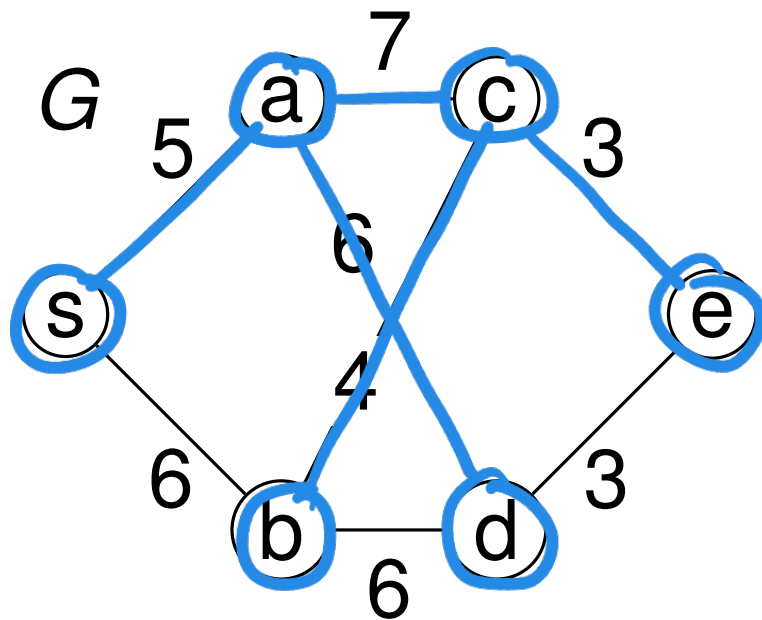
A tree in a graph that uses every
vertex

(d) **minimum cost spanning tree**

A spanning tree with the smallest
weight sum

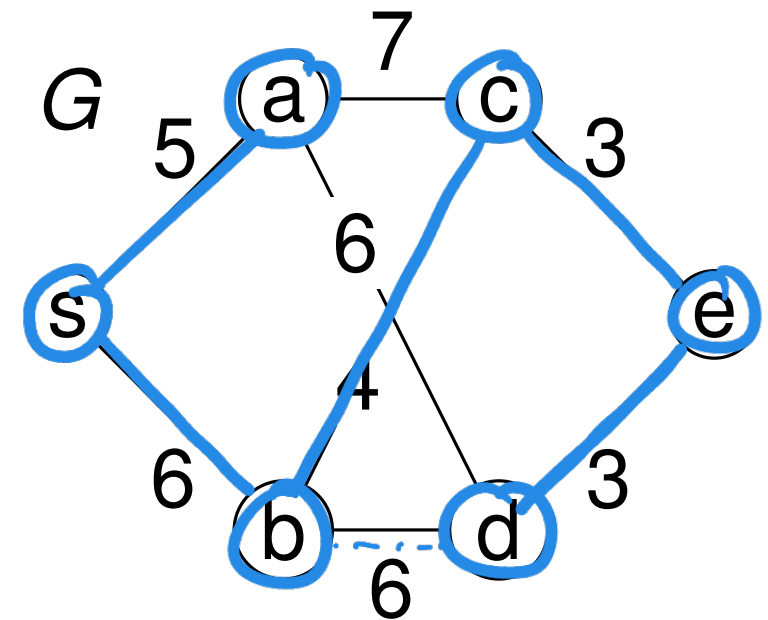


Example:



Weight sum =

$$\begin{array}{ccccc} as & ab & ac & ad & cb \\ 5 & 6 & 7 & 4 & 3 \end{array}$$
$$5 + 6 + 7 + 4 + 3 = 25$$



$$5 + 6 + 4 + 3 + 3 = 21$$

(this is the smallest, but
how can we be sure?)

Kruskal's Algorithm

input: a graph, G , with costs (or weights) on the edges

output: a spanning tree, T , of minimum cost

Steps:

(a) (Initialization Step:) T is a graph on the vertex set of G but with no edges.

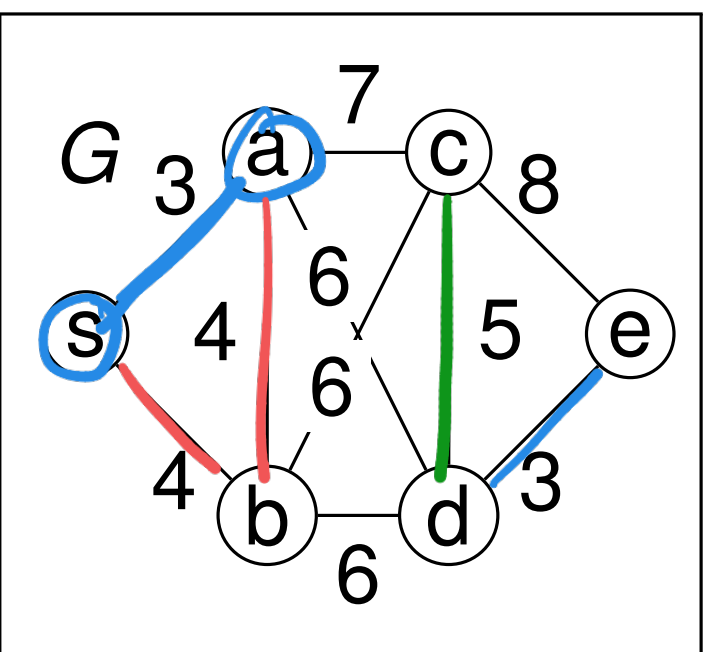
(b) (Iterative Step:)



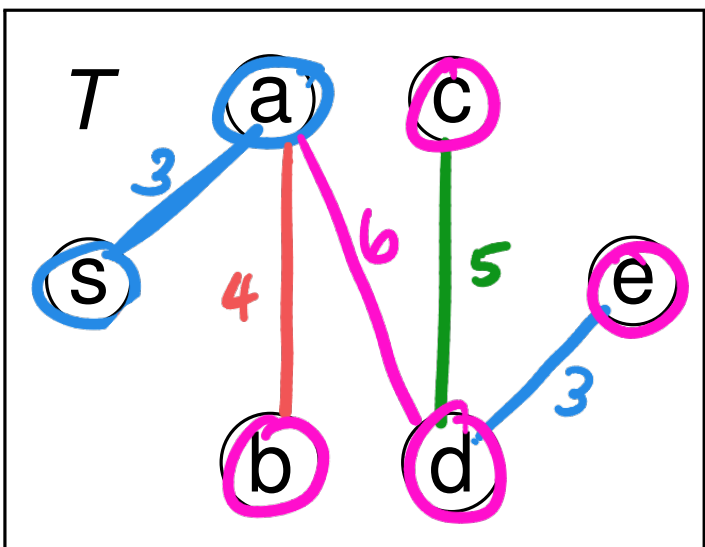
- (i) Select the cheapest unused edge in the graph. (Ties are broken alphabetically.)
- (ii) If the edge does **not** create a cycle, add the edge to T . Otherwise, reject the edge.
- (iii) Mark the edge as used.
- (iv) If T is a spanning tree, **STOP**. Otherwise return to the beginning of the iterative step.

Use Kruskal's Algorithm

Use Kruskal's Algorithm to find the minimum cost spanning tree for the graph G below.



total weight for
our MCST =
 $3 + 4 + 6 + 5 + 3 = 21$



Used?	edges	weights
✓	as	3
✓	dc	5
✓	ab	4
✗	bs	4
	cd	8
	ad	6
	bc	6
	bd	6
	ac	7
	ce	8

STOP! We have a spanning tree!

Think of an application of Kruskal's Algorithm.

(where might you want a minimum cost spanning tree?)

- cities & roads
- houses & connections to water main
- webpages & links
fewest # of links to connect all the web pages!